

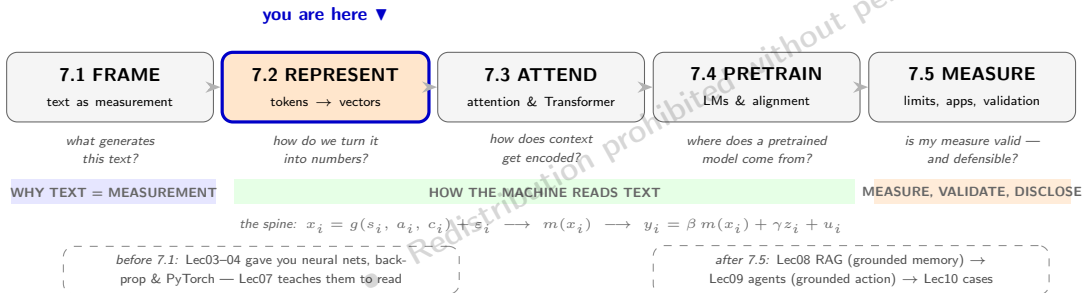
AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.2b: Word Embeddings

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Lecture 7 in One Map



Route: from counts to coordinates (part 2/2 of REPRESENT) — Word2Vec and GloVe, the geometry of meaning, what a direction in the space can measure, and where static vectors break.

From Counts to Coordinates

Learning a vector space in which meaning has geometry

Why Embed Tokens into Vectors?

- Neural networks operate on real-valued vectors, not discrete symbols.
- An **embedding** is a learned function

$$\mathcal{E}: \{1, \dots, M\} \longrightarrow \mathbb{R}^d,$$

mapping each token ID to a dense vector of dimension d (typically $d = 128$ to 4096).

- **What we want from embeddings:**

- Semantically similar tokens should have nearby vectors (close coordinates in the learned basis)
- Algebraic operations (sums, differences) should reflect semantic relations
- The mapping should be *learned* from data, not hand-designed

- **Bonus:** we will also need a *positional* signal so the model knows token order—more on this when we meet the Transformer (Act 3).

Three Approaches to Word Representation

The problem from the previous slide: find a mapping from tokens to vectors that captures semantic relationships. Three generations of methods have tried to solve this, each fixing a specific failure of its predecessor. Understanding this progression explains *why* the Transformer was needed.

Method	Type	Idea	Year
TF-IDF	Sparse, frequency	Re-weight word counts	1970s
Word2Vec	Dense, predictive	Predict neighbors in window	2013
GloVe	Dense, statistical	Factor co-occurrence matrix	2014
Transformer embeddings	Dense, contextual	Vary by surrounding tokens	2017+

Word2Vec: Dense Embeddings from Context

- **Distributional hypothesis** (Firth, 1957): “*You shall know a word by the company it keeps.*”
- **Idea** (Mikolov et al., 2013): for each word in a corpus, look at a window of $\pm L$ neighboring words. Learn a vector for the center word that *predicts* its neighbors (*Skip-gram*) or learn one that *is predicted by* its neighbors (*CBOW*).
- **Training output:** a single static vector $\mathbf{u}_w \in \mathbb{R}^d$ for each word w in the vocabulary, with d typically 100–300.
- **Geometric magic:** after training, simple vector arithmetic captures analogies, e.g.

$$\mathbf{u}_{\text{king}} - \mathbf{u}_{\text{man}} + \mathbf{u}_{\text{woman}} \approx \mathbf{u}_{\text{queen}}.$$

Words living on the same semantic axis differ by approximately the same vector.

Skip-Gram in One Slide: The Math

- Each word w has two learned vectors: an input vector $\mathbf{u}_w$ and an output vector $\mathbf{v}_w$ (often merged after training).
- For a center word w_t and a context word w_c in its window, the model predicts

$$P(w_c | w_t) = \frac{\exp(\mathbf{u}_{w_t}^\top \mathbf{v}_{w_c})}{\sum_{j=1}^M \exp(\mathbf{u}_{w_t}^\top \mathbf{v}_{w_j})}.$$

- Training objective (negative log-likelihood over all (center, context) pairs):

$$\mathcal{L} = - \sum_t \sum_{c \in \text{window}(t)} \log P(w_c | w_t).$$

- Negative sampling (SGNS)** makes this tractable — instead of the full-vocabulary softmax, push up the dot product of each *true* (center, context) pair and push down $K \in [5, 20]$ randomly drawn pairs:

$$\mathcal{L}_{\text{SGNS}} = \log \sigma(\mathbf{v}_c^\top \mathbf{u}_t) + \sum_{k=1}^K \mathbb{E}_{n_k \sim P_{\text{noise}}} [\log \sigma(-\mathbf{v}_{n_k}^\top \mathbf{u}_t)].$$

“inflation” near “persistent”: pushed together; “inflation” vs. “basketball”: pushed apart.

Skip-Gram: Walkthrough on “central banks target inflation”

- **Sentence:** “central banks target inflation”, window radius $L = 2$, center word = *target*.
- Training pairs formed from this center word:

$(target, central), (target, banks), (target, inflation).$

- For each pair, the score is the dot product $\mathbf{u}_{target}^T \mathbf{v}_{context}$. Gradient descent *increases* this score for observed pairs and *decreases* it for random (negative) pairs.
- **Across the whole corpus:** *target* appears near *rates, inflation, policy, Fed*. So do *raise* and *cut*. Consequently $\mathbf{u}_{target}$ ends up close to $\mathbf{u}_{raise}, \mathbf{u}_{cut}$ and far from $\mathbf{u}_{apple}$.
- **Takeaway:** analogies emerge because consistent co-occurrences place related words at similar positions in $\mathbb{R}^d$.

GloVe: A Gravity Equation for Words

- Pennington et al. (2014): don't slide a window and predict — **count once, then fit**. Let X_{ij} = how often words i and j appear near each other in the *whole* corpus:

$$\underbrace{\mathbf{u}_{w_i}^\top \mathbf{v}_{w_j}}_{\text{affinity of the pair}} + \underbrace{b_i + b_j}_{\text{word "fixed effects"}} \approx \underbrace{\log X_{ij}}_{\text{observed togetherness}}$$

- Read it like an economist — it is a gravity equation.** In trade:

$\log(\text{trade}_{ij}) \approx \text{FE}_i + \text{FE}_j + \text{pair relationship}$. Here:

- $\log X_{ij}$: how much the two words “trade” with each other. The log matters because counts span orders of magnitude (3 vs. 820) — same reason we work with log GDP.
- b_i, b_j : word fixed effects — *the* co-occurs with everything simply because it is everywhere; the fixed effects absorb that baseline popularity.
- $\mathbf{u}_{w_i}^\top \mathbf{v}_{w_j}$: what is left over — the pair's *specific affinity*, net of how common each word is. This is the only part that becomes “meaning.”
- So the fit forces:** words that keep each other's company (after controlling for popularity) get vectors pointing the same way; words that never meet stay at right angles. Meaning becomes geometry, estimated like a regression.

GloVe: The Equation at Work on a Central-Bank Corpus

Toy co-occurrence counts X_{ij} (illustrative):

	<i>rates</i>	<i>inflation</i>	<i>apple</i>
Fed	820	410	3
ECB	540	390	2
hawkish	210	160	0

What the fit does, row by row:

- *Fed-rates*: 820 co-occurrences $\Rightarrow \log X \approx 6.7$. Only a **large dot product** can produce that $\Rightarrow \mathbf{u}_{\text{Fed}}$ is pulled into alignment with $\mathbf{v}_{\text{rates}}$.
- *Fed-apple*: 3 co-occurrences $\Rightarrow \log X \approx 1.1$ — the fixed effects alone explain that much $\Rightarrow$ no affinity left to fit $\Rightarrow$ **near-orthogonal** vectors.
- *hawkish*: keeps the same company as Fed and ECB (*rates, inflation, never apple*) $\Rightarrow$ pulled into the **same region** of the space.

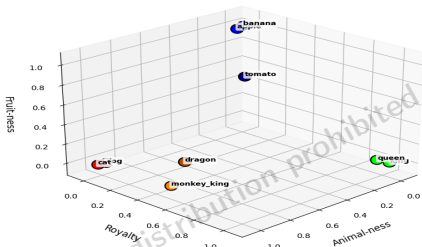
The punchline: nobody told the model what a central bank is — rows with similar co-occurrence patterns are forced to similar coordinates, and “monetary-policy vocabulary” emerges purely from counting. (Word2Vec reaches the same geometry by streaming prediction instead of one global fit.)

The Geometry of Meaning

Distances, directions, analogies — and the biases baked in

A 3D Toy of Word Embeddings

Enhanced 3D Word Embeddings Visualization



The three axes loosely correspond to “Animal-ness”, “Royalty”, and “Fruit-ness”. Words like *cat*, *king*, and *apple* cluster along the dimension that fits them best. Real LLM embeddings live in hundreds of dimensions, but the same intuition holds.

We compare words by the *direction* of their vectors — **cosine similarity** $\cos(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ — not raw distance: frequent words grow longer vectors in training, and direction strips that out.

Analogies and the Bias Problem

- **Geometric magic** (Mikolov et al., 2013): simple vector arithmetic captures analogies.

$$\mathbf{u}_{\text{king}} - \mathbf{u}_{\text{man}} + \mathbf{u}_{\text{woman}} \approx \mathbf{u}_{\text{queen}}$$

$$\mathbf{u}_{\text{Paris}} - \mathbf{u}_{\text{France}} + \mathbf{u}_{\text{Italy}} \approx \mathbf{u}_{\text{Rome}}$$

- Words living on the same semantic axis differ by approximately the same vector — gender, capital city, comparative tense, monetary stance. (Caveat: the clean linear pattern weakens for specialized, low-frequency vocabulary — validate on your own terms, don't extrapolate from benchmark word lists.)
- **Bias is encoded too** (Bolukbasi et al., 2016, *NeurIPS*; Caliskan, Bryson & Narayanan, 2017, *Science*):
 - $\mathbf{u}_{\text{man}} - \mathbf{u}_{\text{woman}}$ has a large component on *programmer* vs. *homemaker*.
 - Word embeddings reproduce IAT-style biases at human magnitudes.
 - For economic text: occupation/gender, race/credit-risk, region/sentiment correlations all transfer from the training corpus into the geometry.
- **Lesson:** embeddings are a mirror of the corpus. Validating that a sentiment vector reflects *policy* stance and not *speaker identity* is a real research step (returns in T5b). Flip side: the encoded

Four Generations of Representation, One Table

Every step toward “understanding” trades away interpretability and economy.

	Counts	TF-IDF	Word2Vec	Contextual (BERT)
Dimension	$V \sim 10^4\text{--}10^5$	same	$d = 100\text{--}300$	$d = 768\text{--}1024$
Sparsity	>99% zeros	>99% zeros	dense	dense
Semantic similarity	none (orthogonal)	weak (IDF re-weight)	yes (distributional)	strong (context-aware)
Polysemy	no	no	one static vector	varies with context
Interpretability	high (word = axis)	high	low (latent space)	very low
Data needed	none	none	$> 10^8$ tokens	$> 10^{10}$ tokens
Compute	trivial	trivial	moderate	high
Typical econ use	EPU, dictionaries	retrieval, similarity	product networks, culture scores	sentiment, stance, extraction

Rule: when external validity rests on an auditable, line-by-line evidence chain, the fancier representation is often the *worse* choice. Pick the rung your research design needs — not the newest one.

From Analogies to Constructs: The Hawk–Dove Direction (Thread B)

Define an economic concept as a direction in the space.

- Candidate monetary-stance axis: $\hat{\mathbf{v}} = \mathbf{u}_{\text{hike}} - \mathbf{u}_{\text{cut}}$. Score any FOMC passage by the cosine of its (averaged) vector with $\hat{\mathbf{v}}$.
- The same trick, generalized: culture scores from seed-word directions (Li, Mai, Shen & Yan 2021); time-varying product-market networks from 10-K business-description similarity (Hoberg & Phillips 2016).
- **Validation idea:** a genuine stance direction should co-move with policy — check the $\hat{\mathbf{v}}$ -scores against subsequent fed-funds-rate changes before using them in a regression.

The separability assumption. *king* – *man* + *woman* works because the difference is (mostly) one clean dimension — everything else cancels. But *hike* co-occurs with overheating and *cut* with recessions, so $\hat{\mathbf{v}}$ may encode **stance and the business cycle at once** — like a bond that bundles duration and credit risk in one price. Purge or control for the cycle dimension before calling it “policy stance”; ask which institutional features make the coupling worse.

Limits and Practice

Where static vectors break — and how to train useful ones anyway

Limits of Static Embeddings

What static embeddings miss

- **One word, many meanings (polysemy):** “rate” = interest rate? exchange rate? growth rate? One vector for all senses.
- **Word order ignored:** “GDP growth NOT slowing” and “GDP growth slowing” have nearly identical mean vectors.
- **Long-range context:** fixed window misses dependencies across paragraphs.
- **Meaning drifts over time:** “mask” before vs. after 2020 are very different concepts.

What we need next

- Vectors that change with *context*, not just word identity
- A mechanism for tokens to “look at” other tokens, near or far
- A way to learn arbitrary relational patterns, not just co-occurrence within a window

⇒ **The Transformer (Act 3).**

Text Series in a VAR: Two Traps

Before a text measure enters a time-series model, clear two hurdles specific to language data.

- **Trap 1 — look-ahead in the weights (recall T2a):** full-sample IDF weights or full-sample embeddings leak future documents into period- t regressors; historical fit becomes unreproducible in real time. Rolling re-estimation (the Word2Prices pattern, T5b) is the fix.
- **Trap 2 — language regime breaks:** a crisis changes how journalists use “liquidity”, “contagion”, “risk”. A structural break in the text series may be a break in *reporting language*, not in the economy — and Bai–Perron/Chow tests cannot tell the two apart.
 - Diagnostics: check whether word neighborhoods are stable across subperiods; estimate by segment; add regime dummies.
 - Plus the usual discipline: stationarity tests on text series, like any other regressor.

The corpus, time span, and preprocessing pipeline are the *sample description* of a text variable — report them as such, or the series cannot be replicated.

Training Your Own: Window, Dimension, Corpus

Three practical dials — and the one that dominates.

- **Window** L : small ($L \leq 3$) $\rightarrow$ syntactic collocations (“quantitative” next to “easing”); large ($L \geq 10$) $\rightarrow$ topical neighborhoods (“QE” near “liquidity injection”, “asset purchases”). Economic measurement usually wants the topical setting.
- **Dimension** d : 100–300 is standard; dense policy vocabularies may need the top of that range to separate fine-grained terms.
- **Corpus domain — the dominant dial**: generic vectors put *discount* near retail sales and *yield* near harvests. On 10-Ks or FOMC minutes, that is a mis-measured proxy with unknowable bias direction — and more data from the wrong domain does not fix it. Retrain (or fine-tune) on the target corpus (Gentzkow, Kelly & Taddy 2019).
- **Chinese policy corpora to train on**: CNRDS filings and MD&A sections; the PBOC *Monetary Policy Report* (quarterly since 2001); Wind financial news. Then the Li et al. recipe transfers directly: seed words $\rightarrow$ nearest-neighbor dictionary expansion $\rightarrow$ firm-level scores — keeping T1’s SOE/non-SOE validity split in view.

Bridge: One Vector per Word Is Not Enough

Static embeddings solved orthogonality — and hit three walls.

- *rate* the policy instrument vs. *rate* the labor-market statistic: one vector must average every sense.
- Averaging kills negation: “growth is *not* slowing” $\approx$ “growth is slowing” in mean-vector space — yet that “not” is the forward guidance.
- Meaning moves with corpus and period, but the vector is frozen at training time.
- **Act 3 fixes all three with one idea:** compute a *fresh* vector for every occurrence, from its actual context — self-attention. First the mechanism (T3a), then the full architecture (T3b).

Arc: T1 FRAME $\rightarrow$ T2 REPRESENT $\rightarrow$ T3 ATTEND $\rightarrow$ T4 PRETRAIN $\rightarrow$ T5 MEASURE.